

Title: Strength of Materials

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Chapter 4: Bending Stress in Beams

When a beam is subjected to a bending moment M , the bending equation $\sigma / y = M / I = E / R$ relates the bending stress σ at a distance y from the neutral axis to the moment of inertia I and the modulus of elasticity E .

Example 4.3: A simply supported beam of length 4 m carries a point load of 20 kN at mid-span. The cross-section is rectangular, 100 mm wide and 200 mm deep. Determine the maximum bending stress $\sigma_{\max}$ in N/mm² (MPa), given that the bending moment $M = W L / 4$.

Solution: $M = 20 \text{ kN} \cdot 4 \text{ m} / 4 = 20 \text{ kN}\cdot\text{m}$. $I = b d^3 / 12 = 100 \cdot 200^3 / 12 \text{ mm}^4$. $\sigma_{\text{max}} = M y / I$ where $y = d / 2 = 100 \text{ mm}$.

Problem 4.12: Repeat the above calculation for a cantilever beam of length 2 m carrying a point load of 10 kN at the free end, with the same rectangular cross-section.

Chapter 5: Shear Force and Torsion

The shear stress τ in a circular shaft of radius R subjected to torque T is $\tau = T r / J$, where J is the polar moment of inertia and r is the radial distance from the shaft axis, measured in N/mm^2 .